

“Genomic And Phenotypic Characterization of Malting Barley and Naked Multi-use Barley as Winter
Crops for New York State”
by Karl Kunze

Barley (*Hordeum vulgare* L.) was one of the first domesticated crops in the ancient world. Barley has been the main source of grain for the working class for millennia, and has found many uses as food, animal feed and malt for brewing and distilling. Increasing consumer preferences for local and organically produced products have brought barley back into the spotlight for multiple uses. Naked barley, where the hull threshes freely from the grain, has had increased attention as the result of higher demand for whole grain products. The demand for beer and its ingredients to be produced locally have incentivized barley to be grown in nontraditional environments. Characterization of barley in novel environments provides many opportunities for local agriculture. My goals were to understand and characterize both winter malting and organic naked barley for the New York State environment and across the country. Despite both areas of research focusing on barley improvement, many of the target goals for winter malting barley breeding varied from the goals of organic multi-use barley breeding. The projects of my dissertation were separated by organic naked barley and winter malting barley. In areas where research methods overlapped, I gained great insight applying research methods across both projects, however most of my research presented here relates to separate goals for each project. Specific goals included 1) Assess the effect of genotype by environment interactions on winter naked barley across the northern United States 2) Use genome wide association in an a naked barley diversity panel to identify quantitative trait loci under organic conditions 4) Identify methods of measurement for barley and weed interaction in an organic breeding context by using visual assessment and high throughput aerial imaging 4) study the genetic relationships between preharvest sprouting (PHS), seed dormancy, and seed germination to identify winter barley germplasm with high malting quality and PHS resistance and 5) develop a winter malting barley breeding pipeline for non-traditional environments.